Week 8 Solution

NPTEL PSOSM Course (Privacy and Security in Online Social Media)

Question 1

What are the different approaches to profile linking? (multiple correct)

(a) Study and analyse common attributes (place of work, profile picture, name, etc.) across different social media platforms ✓

(b) Check important attributes such as the same emails, locations, and contact numbers (if available) used across multiple platforms ✓

(c) Only looking at the user posting behaviour on social media platforms will help in profile linking.

(d) It is not possible to find the same user profiles on different social media platforms.

Solution

Profile linking (also called identity resolution) means finding the accounts that belong to the same real person on different platforms, for example a person's Twitter, Facebook and LinkedIn profiles.

- **(a) Common profile attributes:** name, username, profile picture, place of work, bio and so on are compared across platforms. Similar values strongly suggest the same person.
- **(b) Unique identifiers:** an email ID or phone number reused across platforms is a near-certain match, and location adds supporting evidence.
- **(c) is wrong** because of the word 'only'. Posting behaviour (writing style, posting times) can help, but it is one signal among many, not the sole approach.
- **(d) is wrong:** profile linking is possible and is done routinely, which is exactly why it is a privacy concern.

Answer: Options (a) and (b).

Question 2

Which of the following statements is/are true regarding username change on social media?

S1: Changes to any profile attribute other than username do not lead to unreachability to the user profile.

S2: The potential benign reasons for username change include space gain, suiting a trending event, gaining/losing anonymity, adjusting to real-life events, and avoiding boredom.

S3: The malicious intentions for username change may include obscured username promotion and username squatting.

S4: Search by the user's old username results in a failed attempt to reach the user's profile, potentially making others falsely believe that the user account has been deactivated.

(a) Only S1 and S4 are correct

(b) Only S1, S2, and S3 are correct

(c) S1, S2, and S4 are correct

(d) All S1, S2, S3, S4 are correct. ✓

Solution

On Twitter, the profile URL is built from the username (twitter.com/username). This makes username special among profile attributes. Checking each statement:

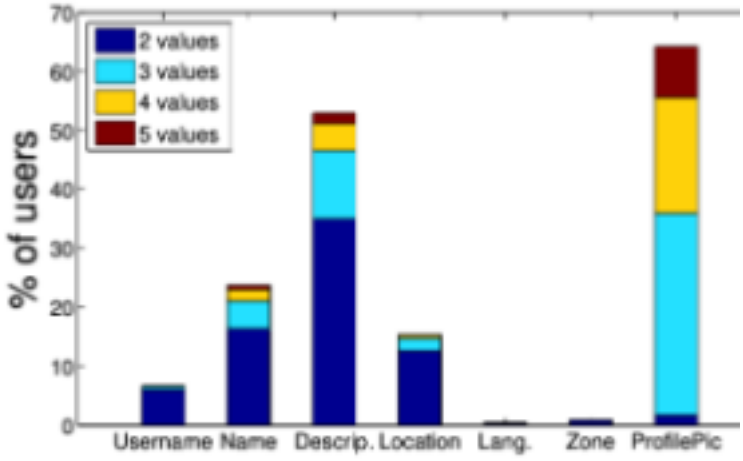
- **S1 (True):** changing the name, bio, location or picture leaves the URL unchanged, so the profile stays reachable. Only a username change breaks the link.

- **S2 (True):** these are the benign reasons identified in the lecture's study: freeing up characters in a tweet (space gain), joining a trend or event, becoming more or less anonymous, real-life changes (marriage, new job) and simply boredom.
- **S3 (True):** malicious users change usernames to promote a hidden or misleading handle, or to squat on a popular username, holding it to impersonate someone or to sell it later.
- **S4 (True):** after a change, the old URL and old @mentions lead nowhere, so friends may wrongly think the account was deleted or suspended. This is the unreachability problem.

Answer: All S1, S2, S3, S4 are correct.

Question 3

Below is the graph of username change. Which of the following is the correct inference from the graph (as discussed in the lecture)?



(a) Around 35 per cent of the users change their profile picture at least 3 times. ✓

(b) Around 60-70 percent of the users change their usernames at least once.

(c) Around 35 percent of the users change their profile picture 3 times.

(d) Around 50-55 per cent of users have changed their descriptions thrice.

Solution

How to read the chart: each bar is a profile attribute. The total height of a bar is the percentage of users who changed that attribute at least once. The coloured segments split those users by how many distinct values the attribute took (2, 3, 4 or 5 values).

Key step: an attribute with k distinct values was changed k - 1 times. So 2 values = 1 change, 3 values = 2 changes, 4 values = 3 changes, 5 values = 4 changes.

Approximate readings from the chart:

Attribute	2 values	3 values	4 values	5 values	Total height
Username	~6%	~0%	~0%	~0%	~6%
Description	~35%	~12%	~3%	~2%	~52%
Profile picture	~2%	~34%	~19%	~10%	~65%

Checking each option:

- **(a) True:** changing the profile picture at least 3 times means 4 or more values: the yellow (4 values) plus dark red (5 values) segments. These run from about the 35% mark up to the top of the bar at about 65%, which is roughly 30 to 35% of users. This matches 'around 35 per cent'.

- **(b) False:** the username bar is only about 6% tall. The 60 to 70% bar is the profile picture, not the username.
- **(c) False:** changing the picture exactly 3 times means exactly 4 values, which is only the yellow segment, about 19%. The light-blue segment of about 34% is 3 values, which is 2 changes, not 3.
- **(d) False:** about 52% is the total description bar (changed at least once). Changing the description thrice means 4 values, which is only about 3%.

Answer: Around 35 per cent of the users change their profile picture at least 3 times.

Question 4

Why do Whisper users have a low clustering coefficient over the network?

(a) Whisper users are likely to interact with complete strangers who are highly unlikely to interact with each other. ✓

(b) Users interact with a large sample of other users

(c) New users make 20% of the contribution in the Whisper content.

(d) All of the above

Solution

The local clustering coefficient of a node measures how many of its neighbours are also connected to each other:

$$C(v) = (\text{number of links among } v\text{'s neighbours}) / (k(k-1)/2), \quad \text{where } k = \text{degree of } v$$

In a friendship network like Facebook, your friends usually know each other (friends of friends are friends), so triangles are common and the coefficient is high.

Whisper is an anonymous app: there are no profiles or friend lists, and users reply to posts by people they do not know. If user v replies to strangers X and Y, there is no reason X and Y should ever interact. Very few triangles form, so the clustering coefficient is low, close to that of a random graph.

Option (b) describes breadth of interaction, not the missing links among neighbours, and option (c) is unrelated to network structure.

Answer: Whisper users are likely to interact with complete strangers who are highly unlikely to interact with each other.

Question 5

What is the primary purpose of Gephi software?

(a) Video Editing

(b) 3D Modeling

(c) Network Visualization and Analysis ✓

(d) Music Composition

Solution

Gephi is a free, open-source desktop tool for exploring graphs and networks. It is widely used in social media research to:

- Load network data (nodes and edges, for example a follower graph) from CSV, GEXF or GraphML files.
- Draw the network with layout algorithms such as ForceAtlas2 and Fruchterman-Reingold.
- Compute metrics: degree, betweenness, closeness, eigenvector centrality, clustering coefficient and modularity (community detection).
- Colour and size nodes by these metrics to spot influencers and communities.

Answer: Network Visualization and Analysis.

Question 6

Why is studying username change behaviour useful for social media researchers? (multiple correct)

(a) This is one of the unique attributes of the user ✓

(b) It is usually homogeneous ✓

(c) The number of characters and length of the username is restricted ✓

(d) It is the publicly available attribute ✓

Solution

The lecture explains why username is a good attribute to study:

- **Unique:** no two accounts on a platform can share a username, so it identifies a user exactly.
- **Homogeneous:** usernames follow the same rules and format on a platform (letters, digits and underscore), so they are easy to compare and analyse at scale.
- **Restricted length:** there is a character limit (for example 15 characters on Twitter), so all usernames are short strings of similar form.
- **Publicly available:** anyone can see a username without special permission, so data can be collected for research.

Answer: All four options.

Question 7

In the network graph of Twitter, where nodes are users, the in-degree of the node would be

(a) Number of followers on Twitter ✓

(b) Number of people you follow

(c) Both a and b

(d) None of the above

Solution

Twitter's graph is directed: an edge X → Y means 'X follows Y'. For a user Y:

- **In-degree** = edges coming into Y = number of users who follow Y = followers.
- **Out-degree** = edges going out of Y = number of users Y follows = following.

Option (b) is the out-degree, so (c) is also wrong.

Answer: Number of followers on Twitter.

Question 8

Which of the following is/are anonymous social networks? (multiple correct)

(a) Secret ✓

(b) 4chan ✓

(c) Yik Yak ✓

(d) Blind ✓

Solution

An anonymous social network lets users post without revealing their real identity. All four options are examples:

Platform	How it is anonymous

Secret	App for sharing posts with friends of friends without showing who wrote them
4chan	Image board where most posts are made under the default name 'Anonymous', with no account needed
Yik Yak	Location-based app showing anonymous posts from people nearby, popular on college campuses
Blind	Workplace app where verified employees post anonymously about their company

Whisper (Question 4) belongs to the same category.

Answer: Secret, 4chan, Yik Yak and Blind (all four).

Question 9

A scale-free network can be one having

(a) Power law degree distribution

(b) Zipf's law

(c) Pareto principle

(d) All of the above ✓

Solution

A network is scale-free when the fraction of nodes with degree k follows a power law:

$$P(k) \propto k^{(-\gamma)}, \quad \text{typically } 2 < \gamma < 3$$

Most nodes have very few links while a few hubs have a huge number (for example, a few celebrity accounts with millions of followers). Zipf's law and the Pareto principle are other forms of the same power-law pattern:

- **Zipf's law:** the item of rank r has frequency proportional to 1/r, so ranking nodes by degree gives a power-law curve.
- **Pareto principle (80/20 rule):** about 20% of nodes hold about 80% of the links, which is a consequence of a heavy-tailed power-law distribution.

Answer: All of the above.

Question 10

Imagine you're a student in a university, and you survey your classmates to find the number of friends each has on one popular social media platform (say Facebook).

The average number of friends reported is 300. However, when you examine each classmate's number individually, many of them have fewer than 300 friends, perhaps ranging from 50 to 200. Why did you observe such a phenomenon?

(a) Most of the individuals have reported it wrong in the report.

(b) Few individuals have exceptionally high friend counts (outliers) that skew the average upward. ✓

(c) There is an error in individual calculations. It is impossible to have 300 friends on average if many individuals have fewer than 300 friends.

(d) None of the above

Solution

Friend counts on social media follow a heavy-tailed (power-law) distribution, as in Question 9: most people have a modest number of friends and a few have very many. The mean is very sensitive to such extreme values, while the median is not.

Numerical illustration (10 classmates):

Friend counts	Statistic
50, 80, 100, 120, 150, 150, 170, 180, 200, 1800	Sum = 3000
Mean = 3000 / 10	300
Median = (150 + 150) / 2	150

Nine of the ten students have 200 friends or fewer, yet the mean is 300 because one student has 1800 friends. The data are correct; the outlier pulls the average up. This is a right-skewed distribution, where mean > median.

Why the others are wrong: (a) assumes reporting errors with no evidence; (c) is mathematically false, since an average can exceed most individual values whenever a few values are very large.

Answer: Few individuals have exceptionally high friend counts (outliers) that skew the average upward.

Answer Key

Q.No.	Correct answer
1	(a) and (b): common attributes; same emails, locations, contact numbers
2	All S1, S2, S3, S4 are correct
3	Around 35 per cent of users change their profile picture at least 3 times
4	Users interact with strangers who are unlikely to interact with each other
5	Network Visualization and Analysis
6	All four options
7	Number of followers on Twitter
8	Secret, 4chan, Yik Yak, Blind (all four)
9	All of the above
10	Few outliers skew the average upward